Notes on Quantum Optics - Section 9

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9 Photon sources and their characterization

9.1 Quantum model for photon detection processes

The properties of a light source regard the way the emitted field can interact with matter, and the way it can influence measurements. It is thus opportune, before delving in the discussion of such a topic, to provide a model for the detection process of a quantized light field. We discuss here, in particular, a simple model first derived by Roy Glauber.[1]

Photon detection processes are typically based on the absorption of a photon by some piece of material (e.g. the photocatode in the case of the photo-multiplier tube). In the following we will consider such absorbing material as a point-like two-level system, initially set in its ground state.

From the quantum mechanical point of view, we are interested in studying the *matrix elements* related to the absorption transition.[2] In the Jaynes-Cummings model, we have described the interaction of one mode of the electromagnetic field with a two-level system using the interaction Hamiltonian:

$$\hat{H}_I = -\iota d\mathcal{E}\hat{a}\hat{\sigma}^\dagger + \iota d\mathcal{E}\hat{a}^\dagger\hat{\sigma} = -d\hat{\sigma}^\dagger\hat{E}^+ + d\hat{\sigma}\hat{E}^- \tag{9-1}$$

Here we have used the decomposition of the electric field operator into its positive- and negative-frequency parts $\hat{E}^+$ and $\hat{E}^-$, defined by the following relation:

$$\hat{E} = \hat{E}^+ + \hat{E}^- = \underbrace{\iota\mathcal{E}\hat{a}e^{\iota\vec{k}\cdot\vec{r}}}_{\hat{E}^+}\underbrace{-\iota\mathcal{E}\hat{a}^\dagger e^{-\iota\vec{k}\cdot\vec{r}}}_{\hat{E}^-} \tag{9-2}$$

Taking into account the long-wavelength approximation, in $\hat{H}_I$ one substitutes these expressions with $e^{\iota\vec{k}\cdot\vec{r}} = e^{-\iota\vec{k}\cdot\vec{r}} = 1$.

In case many modes of the electromagnetic field are involved, it is reasonable to keep a similar expression for $\hat{H}_I$, except that:

$$\hat{E}^+ = \iota\sum_l \mathcal{E}_l\hat{a}_l e^{\iota\vec{k}_l\cdot\vec{r}} \qquad\qquad \hat{E}^- = -\iota\sum_l \mathcal{E}_l\hat{a}_l^\dagger e^{-\iota\vec{k}_l\cdot\vec{r}} \tag{9-3}$$

where the sum runs on all the modes involved, and to write the interaction Hamiltonian $\hat{H}_I$ we can still consider $e^{\iota\vec{k}_l\cdot\vec{r}} = e^{-\iota\vec{k}_l\cdot\vec{r}} = 1$.

In the photon absorption process, the two-level system passes from the ground $|g\rangle$ to the excited $|e\rangle$ state, while the field passes from an initial state $|i\rangle$ to a final state $|f\rangle$ with one photon less. The relevant matrix element for this transition is:

$$H_{i\to f} = \langle f|\langle e|\hat{H}_I|g\rangle|i\rangle \tag{9-4}$$

Employing the expression (9-1) for the interaction Hamiltonian, and considering that $\langle e|\hat{\sigma}^\dagger|g\rangle = 1$ while $\langle e|\hat{\sigma}|g\rangle = 0$, one gets:

$$H_{i\to f} \propto \langle f|\hat{E}^+|i\rangle \tag{9-5}$$

The transition probability per unit time is proportional to the modulus squared of the matrix element:

$$p_{i\to f} \propto |\langle f|\hat{E}^+|i\rangle|^2 = \langle i|\hat{E}^-|f\rangle\langle f|\hat{E}^+|i\rangle \tag{9-6}$$

If a single mode is involved and the initial state of the field is a Fock state $|i\rangle \equiv |n\rangle$, the only relevant final state is the Fock state of the same mode with one photon less $|f\rangle \equiv |n-1\rangle$. However, in the

[1] See R. Glauber, "The Quantum Theory of Optical Coherence" *Phys. Rev.* **130**, 2529 (1963).

[2] We briefly remind the definition of the *matrix element* of the Hamiltonian. In general, we can express the state of a physical system as $|\psi\rangle = \sum_i c_i|u_i\rangle$ where $|u_i\rangle$ are kets forming a complete basis of the Hilbert space, and c_i are complex coefficients. The state is thus completely described by the complex vector $\underline{c} = \{c_i\}$. The Schroedinger equation governs the evolution of the state according to $\hat{H}|\psi\rangle = \iota\hbar\frac{d}{dt}|\psi\rangle$; this can be written for the complex vector $\underline{c}$ as $H\underline{c} = \iota\hbar\frac{d}{dt}\underline{c}$, where $H = \{H_{m,n}\}$ with matrix elements $H_{m,n} = \langle u_m|\hat{H}|u_n\rangle$. It is apparent that the transition from the state $|u_i\rangle$ to the state $|u_j\rangle$ is basically governed by the matrix element $H_{i,j}$.

general case, the initial state of the field is a superposition of many Fock states and involving different modes, thus there could be many different possible final states $|f\rangle$ on which the field could end up to be after the transition. The overall single-photon absorption probability is the sum of the probabilities calculated on all the final states $|f\rangle$:

$$w_1 = \sum_f p_{i\to f} = \sum_f |\langle f|\hat{E}^+|i\rangle|^2 = \sum_f \langle i|\hat{E}^-|f\rangle\langle f|\hat{E}^+|i\rangle \tag{9-7}$$

It is useful to observe that in Eq. (9-7) the sum can be operated not only on those specific states $|f\rangle$ that are reasonably accessible from $|i\rangle$ upon absorption of one photon, but on an entire, complete basis of states $|f\rangle$ of the field. Namely, we run the sum on a set of states $|f\rangle$ such that $\sum_f |f\rangle\langle f| = \mathbb{I}$. Indeed, if some state is not accessible, simply its correspondent term $|\langle f|\hat{E}^+|i\rangle|^2$ is zero. We can thus write:

$$w_1 \propto \sum_f \langle i|\hat{E}^-|f\rangle\langle f|\hat{E}^+|i\rangle = \langle i|\hat{E}^-\left(\sum_f |f\rangle\langle f|\right)\hat{E}^+|i\rangle = \langle i|\hat{E}^-\hat{E}^+|i\rangle \tag{9-8}$$

In conclusion, the single photon absorption probability is proportional to the expectation value of $\hat{E}^-\hat{E}^+$, evaluated on the actual initial state of the field $|i\rangle$. Note that, if a single mode of the field is involved, $\hat{E}^-\hat{E}^+ \propto \hat{a}^\dagger\hat{a} = \hat{n}$.
With a similar reasoning one can develop operators for the simultaneous detection of two photons (we should consider an absorber composed of two independent two-level systems). We would start from matrix elements with the form $\langle f|\hat{E}^+\hat{E}^+|i\rangle$ and end up in a two-photon absorption probability:

$$w_2 \propto \sum_f |\langle f|\hat{E}^+\hat{E}^+|i\rangle|^2 = \langle i|\hat{E}^-\hat{E}^-\hat{E}^+\hat{E}^+|i\rangle \tag{9-9}$$

In the most general case, the probability to detect n photons respectively at the positions $\vec{r}_1 \dots \vec{r}_n$ and at the respective time instants $t_1 \dots t_n$ results to be proportional to the function:

$$G^{(n)}(x_1,\dots,x_n;x_n,\dots,x_1) = \langle \hat{E}^-(x_1)\dots\hat{E}^-(x_n)\hat{E}^+(x_n)\dots\hat{E}^+(x_1)\rangle \tag{9-10}$$

where the expectation value is taken on the relevant state of the field, and we have indicated the spatio-temporal coordinates with the symbol $x = (\vec{r}, t)$. The function $G^{(n)}(x_1,\dots,x_n;x_n,\dots,x_1)$ is called **quantum correlation function of order** n, here evaluated in the set of spatio-temporal coordinates $(x_1,\dots,x_n;x_n,\dots,x_1)$.
We see from Eqs. (9-8) and (9-9) that the probabilities to detect one or two photons (simultaneously) at a point in space and time $x = (\vec{r}, t)$ are just:

$$w_1(x) \propto G^{(1)}(x;x) \qquad\qquad w_2(x) \propto G^{(2)}(x,x;x,x) \tag{9-11}$$

The light intensity detected by a classical detector (such as an optical power-meter or a photodiode) is also proportional to w_1:

$$I(x) \propto G^{(1)}(x;x) \tag{9-12}$$

9.2 First-order coherence

An important concept in optics is that of *coherence* of an optical beam or source. In a classical framework, this concept is inextricably related to the capability of this beam to show interference effects. It is interesting to study how the properties of a state of the electromagnetic field can influence the visibility of interference fringes, within a quantum framework.
Let us consider a Mach-Zehnder interferometer, fed with an arbitrary light state $|\psi\rangle_a$ at one input, and the vacuum $|0\rangle_b$ at the other input. We have studied already the Mach-Zehnder model in Section 6, in case only two optical modes are involved (one per each input port) at the same oscillation frequency

ω. Here we would like to be more general, and we consider that at each input port multi-mode fields may be present, in particular involving different oscillation frequencies ω.
In the practical realization, the internal phase shift ϕ of the Mach-Zehnder is often produced by a difference in length ΔL of the optical paths of the two arms. For each mode l, characterized by its frequency ω_l and its wavevector k_l, one can thus write:

$$\Delta\phi = k_l\,\Delta L = \frac{\omega_l}{c}\Delta L = \omega_l \tau \tag{9-13}$$

where $\tau = \frac{\Delta L}{c}$ is the equivalent temporal delay induced by the length difference. Importantly, the phase shift ϕ depends on ω_l. The Mach-Zehnder interferometer thus acts on each mode with a different matrix, that in this case it will be convenient to write in the following form:

$$M^{(l)}_{\text{MZI}} = \frac{1}{2}\begin{bmatrix} 1-e^{\iota\omega_l\tau} & \iota(1+e^{\iota\omega_l\tau}) \\ \iota(1+e^{\iota\omega_l\tau}) & e^{\iota\omega_l\tau}-1 \end{bmatrix} \tag{9-14}$$

Considering a multi-mode field, the electric field operators related to the output port b' at time t, in the Heisenberg picture, are:

$$\hat{E}^+_{b'}(t) = \iota\sum_l \mathcal{E}_l\,\hat{b}'_l(t) \qquad\qquad \hat{E}^-_{b'}(t) = -\iota\sum_l \mathcal{E}_l\,\hat{b}'^{\dagger}_l(t) \tag{9-15}$$

Here we have dropped the spatial dependency because it is implicit in the mode indication; specification of the time instants is instead relevant.
The output intensity measured at port b' is proportional to:

$$I_{b'}(t) \propto G^{(1)}_{b'}(t;t) = \langle \hat{E}^-_{b'}(t)\,\hat{E}^+_{b'}(t)\rangle = \langle\left(\sum_l \mathcal{E}_l\,\hat{b}'^{\dagger}_l(t)\right)\left(\sum_l \mathcal{E}_l\,\hat{b}'_{l'}(t)\right)\rangle = \sum_{l,l'}\mathcal{E}_l\mathcal{E}_{l'}\,\langle\hat{b}'^{\dagger}_l(t)\hat{b}'_{l'}(t)\rangle \tag{9-16}$$

Using the transformation of the operators according to the Mach-Zehnder matrix (9-14) we obtain:

$$\begin{aligned} G^{(1)}_{b'}(t,t) &= \sum_{l,l'}\mathcal{E}_l\mathcal{E}_{l'}\,\langle\hat{U}^{\dagger}_{\text{MZI}}\hat{b}^{\dagger}_l\hat{U}_{\text{MZI}}\,\hat{U}^{\dagger}_{\text{MZI}}\hat{b}_{l'}\hat{U}_{\text{MZI}}\rangle = \\ &= \sum_{l,l'}\frac{\mathcal{E}_l\mathcal{E}_{l'}}{4}\langle\left[-\iota\left(1+e^{-\iota\omega_l\tau}\right)\hat{a}^{\dagger}_l + \left(e^{-\iota\omega_l\tau}-1\right)\hat{b}^{\dagger}_l\right]\cdot\left[+\iota\left(1+e^{+\iota\omega_{l'}\tau}\right)\hat{a}_{l'} + \left(e^{\iota\omega_{l'}\tau}-1\right)\hat{b}_{l'}\right]\rangle \end{aligned} \tag{9-17}$$

where we have omitted for simplicity to specify the time dependency of the operators, as they are all evaluated at time t.
In developing further the latter expression, it is useful to remind that the quantum expectation value is carried out on an input state where port b is in the vacuum state. Terms containing $\hat{b}^{\dagger}_l$ or $\hat{b}_{l'}$ can be safely dropped.

$$G^{(1)}_{b'}(t,t) = \sum_{l,l'}\frac{\mathcal{E}_l\mathcal{E}_{l'}}{4}\left(1+e^{-\iota\omega_l\tau}\right)\left(1+e^{+\iota\omega_{l'}\tau}\right)\langle\hat{a}^{\dagger}_l\hat{a}_{l'}\rangle \tag{9-18}$$

The expression in Eq. (9-18) can be divided into four double sums:

$$\begin{aligned} G^{(1)}_{b'}(t,t) = \sum_{l,l'}\frac{\mathcal{E}_l\mathcal{E}_{l'}}{4}\langle\hat{a}^{\dagger}_l\hat{a}_{l'}\rangle + \sum_{l,l'}\frac{\mathcal{E}_l\mathcal{E}_{l'}}{4}e^{\iota(\omega_{l'}-\omega_l)\tau}\langle\hat{a}^{\dagger}_l\hat{a}_{l'}\rangle + \sum_{l,l'}\frac{\mathcal{E}_l\mathcal{E}_{l'}}{4}e^{-\iota\omega_l\tau}\langle\hat{a}^{\dagger}_l\hat{a}_{l'}\rangle + \\ + \sum_{l,l'}\frac{\mathcal{E}_l\mathcal{E}_{l'}}{4}e^{+\iota\omega_{l'}\tau}\langle\hat{a}^{\dagger}_l\hat{a}_{l'}\rangle \end{aligned} \tag{9-19}$$

Being $\hat{a}_l(t-\tau) = \hat{a}_l(t)e^{+\iota\omega_l\tau}$ and $\hat{a}_l^\dagger(t-\tau) = \hat{a}_l^\dagger(t)e^{-\iota\omega_l\tau}$, and taking into account Eq. (9-10), the various terms are rewritten as:

$$\sum_{l,l'} \mathcal{E}_l\mathcal{E}_{l'}\langle\hat{a}_l^\dagger\hat{a}_{l'}\rangle = \langle\hat{E}_a^-(t)\,\hat{E}_a^+(t)\rangle = G_a^{(1)}(t,t) \tag{9-20}$$

$$\sum_{l,l'} \mathcal{E}_l\mathcal{E}_{l'}e^{\iota(\omega_{l'}-\omega_l)\tau}\langle\hat{a}_l^\dagger\hat{a}_{l'}\rangle = \langle\hat{E}_a^-(t-\tau)\,\hat{E}_a^+(t-\tau)\rangle = G_a^{(1)}(t-\tau,t-\tau) \tag{9-21}$$

$$\sum_{l,l'} \mathcal{E}_l\mathcal{E}_{l'}e^{-\iota\omega_l\tau}\langle\hat{a}_l^\dagger\hat{a}_{l'}\rangle = \left[\sum_{l,l'} \mathcal{E}_l\mathcal{E}_{l'}e^{+\iota\omega_{l'}\tau}\langle\hat{a}_l^\dagger\hat{a}_{l'}\rangle\right]^* = \langle\hat{E}_a^-(t-\tau)\,\hat{E}_a^+(t)\rangle = G_a^{(1)}(t-\tau,t) \tag{9-22}$$

We conclude:

$$G_{b'}^{(1)}(t,t) = \frac{1}{4}\left[\,G_a^{(1)}(t,t) + G_a^{(1)}(t-\tau,t-\tau) + 2\Re\left\{G_a^{(1)}(t-\tau,t)\right\}\right] \tag{9-23}$$

Going back to the optical intensities, which are proportional to $G^{(1)}$, we write:

$$I_{b'}(t) = I_a(t) + I_a(t-\tau) + \sqrt{I_a(t)I_a(t-\tau)}\Re\left\{g_a^{(1)}(t-\tau,t)\right\} \tag{9-24}$$

where we have defined the normalized **first-order quantum coherence function**:

$$\boxed{g^{(1)}(t-\tau,t) = \frac{G^{(1)}(t-\tau,t)}{\sqrt{G^{(1)}(t,t)G^{(1)}(t-\tau,t-\tau)}} = \frac{\langle\hat{E}^-(t-\tau)\,\hat{E}^+(t)\rangle}{\sqrt{\langle\hat{E}^-(t)\,\hat{E}^+(t)\rangle\langle\hat{E}^-(t-\tau)\,\hat{E}^+(t-\tau)\rangle}}} \tag{9-25}$$

If the field is *stationary*, which means that its properties do not depend on the specific instant of time considered, only relative differences in time instants are relevant and in the calculations one may consider, if needed, that:

$$\hat{E}^-(t)\,\hat{E}^+(t) = \hat{E}^-(t-\tau)\,\hat{E}^+(t-\tau) \tag{9-26}$$

and also:

$$g^{(1)}(t-\tau,t) = g^{(1)}(t,t+\tau) \tag{9-27}$$

The appearance of interference fringes at the output of the Mach-Zehnder interferometer is indeed related the $g^{(1)}(t-\tau,t)$ and in particular the fringe modulations are manifesting the real part of this function.
In the case of a **single-mode coherent field** $|\alpha\rangle$, one has:

$$\hat{E}^+(t)\,|\alpha\rangle = \iota\mathcal{E}\alpha e^{-\iota\omega t}\,|\alpha\rangle \qquad\qquad \langle\alpha|\,\hat{E}^-(t) = -\,\langle\alpha|\,\iota\mathcal{E}\alpha^* e^{+\iota\omega t} \tag{9-28}$$

and the first-order coherence function is simply:

$$g_{|\alpha\rangle}^{(1)}(t-\tau,t) = e^{-\iota\omega\tau} \tag{9-29}$$

which gives, substituted in Eq. (9-24), the usual cosinusoidal interference fringe as a function of $\phi = \omega\tau$. The fact that $|g^{(1)}(t-\tau,t)| = 1$ indicates full visibility[3] for the interference fringes.
An analogous result is retrieved for a **single-mode Fock state** $|n\rangle$, for which:

$$\langle\hat{E}^-(t)\,\hat{E}^+(t)\rangle = \langle\hat{E}^-(t-\tau)\,\hat{E}^+(t-\tau)\rangle = \mathcal{E}^2 n \tag{9-30}$$

$$\langle\hat{E}^-(t-\tau)\,\hat{E}^+(t)\rangle = \mathcal{E}^2 e^{-\iota\omega\tau}\,\langle n|\,\hat{a}^\dagger\hat{a}\,|n\rangle = \mathcal{E}^2 e^{-\iota\omega\tau} n \tag{9-31}$$

[3]We remind the definition of visibility of interference fringes as:

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

with $I_{\max}$ and $I_{\min}$ being the maximum and minimum values for the intensity at the detector, in a period of the interference fringe.

and thus:

$$g^{(1)}_{|n\rangle}(t-\tau,t) = e^{-\iota\omega\tau} \tag{9-32}$$

We have studied indeed in Section 6 the Mach-Zehnder interferometer fed with a single-photon Fock state $|1\rangle$, which shows interference fringes with the same periodicity and visibility of the classical case. This result for the coherence function indicates that the same fringes are retrieved for a generic $|n\rangle$, with no change in periodicity and still full visibility.
A more complex case is that of a **multi-mode coherent field**:

$$|\vec{\alpha}\rangle = \bigotimes_l |\alpha_l\rangle \tag{9-33}$$

which is a tensor product of coherent states with eigenvalue α_l for each mode l. We refer in the following, in particular, to modes of the same spatial order and polarization, differing however in their frequency ω. This is a suitable description, for instance, for the emission of a laser whose spectrum includes several longitudinal modes. These multi-mode states are still eigenstates of the $\hat{E}^+(t)$ operator (which includes all the annihilation operators of the different modes), with eigenvalue $E_{\vec{\alpha}}(t)$:

$$\hat{E}^+(t)\,|\vec{\alpha}\rangle = \iota\sum_l \mathcal{E}_l\hat{a}_l\,|\vec{\alpha}\rangle = \left(\iota\sum_l \mathcal{E}_l\alpha_l e^{-\iota\omega_l t}\right)|\vec{\alpha}\rangle = E_{\vec{\alpha}}(t)\;|\vec{\alpha}\rangle \tag{9-34}$$

Similarly, for the $\hat{E}^-(t)$ operator, we have:

$$\langle\vec{\alpha}|\,\hat{E}^-(t) = -\sum_l \iota\mathcal{E}_l\,\langle\vec{\alpha}|\,\hat{a}_l^\dagger = \langle\vec{\alpha}|\left(-\iota\sum_l \mathcal{E}_l\alpha_l^* e^{+\iota\omega_l t}\right) = \langle\vec{\alpha}|\,E^*_{\vec{\alpha}}(t) \tag{9-35}$$

The $g^{(1)}$ function can be written as:

$$g^{(1)}_{|\vec{\alpha}\rangle}(t-\tau,t) = \frac{E^*_{\vec{\alpha}}(t-\tau)E_{\vec{\alpha}}(t)}{|E_{\vec{\alpha}}(t-\tau)||E_{\vec{\alpha}}(t)|} \tag{9-36}$$

thus it is unitary in modulus $|g^{(1)}_{|\vec{\alpha}\rangle}| = 1$ for every τ. It is interesting however to further analyze the numerator, which determines the actual shape of the interference fringes:

$$E^*_{\vec{\alpha}}(t-\tau)E_{\vec{\alpha}}(t) = \sum_{l,l'} \mathcal{E}_l\mathcal{E}_{l'}\alpha_l^*\alpha_{l'}e^{\iota(\omega_l-\omega_{l'})t}e^{-\iota\omega_l\tau} \tag{9-37}$$

This expression contains many terms rapidly oscillating in time as $e^{\iota(\omega_l-\omega_{l'})t}$ which indicate the possible beating between the waves at different frequency. However, an experimental intensity measurement cannot have an infinite bandwidth, and reasonably averages over these fast oscillations. If we take the time average of Eq. (9-37) the only terms in the sum that give non-negligible contributions are the ones for $\omega_l = \omega_{l'}$, namely $l = l'$. We obtain:

$$\overline{\langle\hat{E}^-(t-\tau)\,\hat{E}^+(t)\rangle} = \lim_{T\to+\infty}\frac{1}{T}\int_0^T E^*_{\vec{\alpha}}(t-\tau)E_{\vec{\alpha}}(t)dt = \sum_l \mathcal{E}_l^2\alpha_l^*\alpha_l e^{-\iota\omega_l\tau} = \sum_{\omega\in\Omega}|\mathcal{E}(\omega)\alpha(\omega)|^2 e^{-\iota\omega\tau} \tag{9-38}$$

Writing $\mathcal{E}_l = \mathcal{E}(\omega)$ and $\alpha_l = \alpha(\omega)$ allows highlight that $\overline{\langle\hat{E}^-(t-\tau)\,\hat{E}^+(t)\rangle}$ is proportional to the discrete Fourier transform of the power spectrum of the field, if the ω_l with non-vanishing spectral contribution can be seen as elements of a reasonable set of evenly spaced frequencies Ω.

9.3 Second-order coherence

Normalized quantum coherence functions of higher order $g^{(n)}(x_1, \ldots, x_n; x_n, \ldots, x_1)$ can be defined as suitably normalized versions of the quantum correlation functions $G^{(n)}(x_1, \ldots, x_n; x_n, \ldots, x_1)$. For the characterization of non-classical light sources particular importance is attributed to the **normalized second-order coherence function**, evaluated at a fixed spatial position and thus written only as a function of time instants as:

$$\boxed{g^{(2)}(t, t+\tau) = \frac{G^{(2)}(t, t+\tau)}{G^{(1)}(t,t)G^{(1)}(t+\tau, t+\tau)} = \frac{\langle \hat{E}^-(t)\hat{E}^-(t+\tau)\hat{E}^+(t+\tau)\hat{E}^+(t)\rangle}{\langle \hat{E}^-(t+\tau)\hat{E}^+(t+\tau)\rangle\langle \hat{E}^-(t)\hat{E}^+(t)\rangle}} \tag{9-39}$$

This function equals the probability to detect two photons at instants t and $t+\tau$, normalized with respect to the product of two single-photon detection probabilities at t and $t+\tau$. In case of stationary states of the field, properties only depends on τ and not on t, so we can write more compactly $g^{(2)}(\tau) = g^{(2)}(t, t+\tau)$.
It is important to note that in reasonable experimental conditions, if τ is sufficiently large, for the purposes of the calculation of (9-39) the state of the field can be written with good approximation as:

$$|\psi\rangle = |\psi(t)\rangle \otimes |\psi(t+\tau)\rangle \tag{9-40}$$

namely the state of the field in t is uncorrelated from the state of the field in $t+\tau$. The operators $\hat{E}^\pm(t)$ and $\hat{E}^\pm(t+\tau)$ are thus acting separately on the two parts of the state and they are basically commuting. We have:

$$\lim_{\tau\to+\infty} \langle \hat{E}^-(t)\hat{E}^-(t+\tau)\hat{E}^+(t+\tau)\hat{E}^+(t)\rangle = \langle \hat{E}^-(t+\tau)\hat{E}^+(t+\tau)\rangle\langle \hat{E}^-(t)\hat{E}^+(t)\rangle \tag{9-41}$$

and hence:

$$\lim_{\tau\to+\infty} g^{(2)}(\tau) = 1 \tag{9-42}$$

For a **single-mode field**, evaluated in $\vec{r} = 0$, we have $\hat{E}^+(t) = \iota\mathcal{E}\hat{a}(t)$ and $\hat{E}^-(t) = -\iota\mathcal{E}\hat{a}^\dagger(t)$ and, as a consequence:

$$g^{(2)}(\tau) = \frac{\langle \hat{a}^\dagger(t)\hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\hat{a}(t)\rangle}{\langle \hat{a}^\dagger(t+\tau)\hat{a}(t+\tau)\rangle\langle \hat{a}^\dagger(t)\hat{a}(t)\rangle} \tag{9-43}$$

This function is of particular interest in $\tau = 0$, where it reads (we write $\hat{a}(t) = \hat{a}$ and $\hat{a}^\dagger(t) = \hat{a}^\dagger$):

$$g^{(2)}(0) = \frac{\langle \hat{a}^\dagger \overbrace{\hat{a}^\dagger\hat{a}}^{=\hat{a}\hat{a}^\dagger - 1} \hat{a}\rangle}{\langle \hat{a}^\dagger\hat{a}\rangle\langle \hat{a}^\dagger\hat{a}\rangle} = \frac{\langle \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a}\rangle - \langle \hat{a}^\dagger\hat{a}\rangle}{(\langle \hat{a}^\dagger\hat{a}\rangle)^2} = \frac{\langle \hat{n}^2\rangle - \langle \hat{n}\rangle}{(\langle \hat{n}\rangle)^2} \tag{9-44}$$

Reminding that $\langle \Delta n^2\rangle = \langle \hat{n}^2\rangle - (\langle \hat{n}\rangle)^2$, the latter expression also equals:

$$g^{(2)}(0) = 1 + \frac{\langle \Delta n^2\rangle - \langle n\rangle}{(\langle \hat{n}\rangle)^2} \tag{9-45}$$

In the case of a **quasi-classical state** $|\alpha\rangle$, we have indeed $\langle \Delta n^2\rangle = \langle n\rangle = \bar{n} = |\alpha|^2$, and thus:

$$g^{(2)}_{|\alpha\rangle}(0) = 1 \tag{9-46}$$

In the case of a **Fock state** $|n\rangle$ instead $\langle n\rangle = n$ and $\langle \Delta n^2\rangle = 0$, so:

$$g^{(2)}_{|n\rangle}(0) = 1 - \frac{1}{n} \tag{9-47}$$

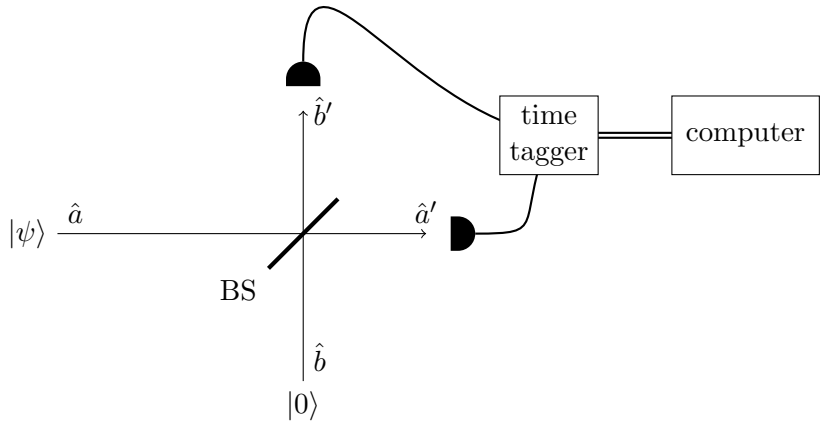

Figure 1: Scheme of an experimental apparatus for measuring the $g^{(2)}$ quantum coherence function for a single-mode in an unknown state $|\psi\rangle$. The state $|\psi\rangle$ is injected in one input of a balanced beam-splitter (BS), whose other input port is in the vacuum state. Two single-photon detector are placed at the two outputs. Subsequent clicks of the two different detectors are collected, measuring the time separation τ between them.

In the particular case of a single-photon Fock state $|n\rangle = |1\rangle$, the second-order coherence function vanishes in $\tau = 0$, as it results:

$$g^{(2)}_{|1\rangle}(0) = 1 - \frac{1}{1} = 0 \tag{9-48}$$

Note that in general, according to Eq. (9-45), the $g^{(2)}$ function in $\tau = 0$ is larger than 1 for a single-mode state with a super-Poissonian statistics and is smaller than 1 for a single-mode state with sub-Poissonian statistics.

The $g^{(2)}(\tau)$ correlation function is measured experimentally with an apparatus similar to the one sketched in Fig. 1. The optical beam in the unknown state $|\psi\rangle$ (assumed as single-mode) is injected in one input port of a balanced beam splitter. The annihilation and creation operators associated to such single mode are $\hat{a}(t)$ and $\hat{a}^\dagger(t)$. There is no light entering in the other input of the beam splitter; namely, the other mode associated to the operators $\hat{b}(t)$ and $\hat{b}^\dagger(t)$ is in the vacuum state $|0\rangle$. Single-photon detectors are placed on the a' and b' output modes. These are used to record pairs of photo-counts that happen in the two different modes (one click per mode) one after the other, within a short time interval τ; the *time tagger* electronics allows to measure τ accurately.

The two output modes are associated respectively to the operators $\hat{a}'$,$\hat{a}'^\dagger$ and $\hat{b}'$,$\hat{b}'^\dagger$, which are evolved operators according to the beam splitter unitary evolution $\hat{U}$ (see Eqs. (7-62) and (7-63)):

$$\hat{a}' = \hat{U}^\dagger \hat{a} \hat{U} = \frac{1}{\sqrt{2}}\left(\hat{a} + \iota \hat{b}\right) \qquad \hat{b}' = \hat{U}^\dagger \hat{b} \hat{U} = \frac{1}{\sqrt{2}}\left(\iota \hat{a} + \hat{b}\right)$$

$$\hat{a}'^\dagger = \hat{U}^\dagger \hat{a}^\dagger \hat{U} = \frac{1}{\sqrt{2}}\left(\hat{a}^\dagger - \iota \hat{b}^\dagger\right) \qquad \hat{b}'^\dagger = \hat{U}^\dagger \hat{b}^\dagger \hat{U} = \frac{1}{\sqrt{2}}\left(-\iota \hat{a}^\dagger + \hat{b}^\dagger\right)$$

We have omitted here to write the time dependency for simplicity of notation.

We indicate as $w_2(\tau)$ the probability to get two detector clicks separated by τ (basically, one click on a at time t and a bit later one click on b at time $t + \tau$; a negative τ means that the b detector is clicking first):

$$w_2(\tau) = \langle\, \hat{a}'^\dagger(t)\, \hat{b}'^\dagger(t+\tau)\, \hat{b}'(t+\tau)\, \hat{a}'(t)\, \rangle = \langle\psi|_a \langle 0|_b\, \hat{a}'^\dagger(t)\, \hat{b}'^\dagger(t+\tau)\, \hat{b}'(t+\tau)\, \hat{a}'(t)\, |\psi\rangle_a |0\rangle_b \tag{9-49}$$

To proceed, we need to take into account that the mode b is in the vacuum state $|0\rangle_b$, hence:

$$\hat{b}\, |0\rangle_b = 0 \qquad \langle 0|_b\, \hat{b}^\dagger = 0 \tag{9-50}$$

Further considering that operators related to different modes commute, the latter condition allows to effectively simplify all occurrences of operators $\hat{b}$ and $\hat{b}^\dagger$ upon substitution of the relations (7-62) and (7-63) in Eq. (9-49):

$$w_2(\tau) = \ldots = \frac{1}{4} \langle\psi|_a \, \hat{a}^\dagger(t)\, \hat{a}^\dagger(t+\tau)\, \hat{a}(t+\tau)\, \hat{a}(t) \, |\psi\rangle_a \tag{9-51}$$

In reasonable experimental conditions, we can consider that for large enough τ operators related to t and to $t+\tau$ become uncorrelated, therefore:

$$w_2(\infty) = \lim_{\tau\to\pm\infty} w_2(\tau) = \frac{1}{4} \langle\, \hat{a}^\dagger(t)\, \hat{a}^\dagger(t+\tau)\, \hat{a}(t+\tau)\, \hat{a}(t)\,\rangle = \langle\, \hat{a}^\dagger(t+\tau)\, \hat{a}(t+\tau)\, \langle\, \hat{a}^\dagger(t)\, \hat{a}(t)\rangle\rangle \tag{9-52}$$

This allows us to estimate experimentally the function $g^{(2)}(\tau)$ as:

$$g^{(2)}(\tau) = \frac{w_2(\tau)}{w_2(\infty)} \tag{9-53}$$

9.4 Single-photon sources

Different non-classical states of light can be employed in the quantum technology applications. Of these, single-photon states are especially useful: namely, we refer to states where one optical mode (or a given combination of modes) is deterministically excited with a single photon. Devices that produce this kind of states are named **single-photon sources**.
More formally, a generic source of light emits wavepackets which may be described by quantum states of the kind:

$$|\psi\rangle = \sum_l \sum_n c_{l,n} \, |n\rangle_l \tag{9-54}$$

The index l runs on all the relevant optical modes. In case the mode decomposition is carried out using plane waves, as in Section 3, l is the usual index condensing the information on the wavevector and on the polarization direction. The ket $|n\rangle_l$ is the Fock state of n photons in the mode l.
If the coefficients $c_{l,n}$ are well determined and are always the same for each successive pulse emitted by the source, the source is indeed emitting a *pure quantum state*, fully described by the expansion in Eq. (9-54). More in general, a source may emit probabilistically a state with a certain sets of $c_{l,n}$, or other ones with different sets of $c_{l,n}$. In this case the emitted quantum state is *mixed* and a more rigorous description should be carried out using the formalism of the *density matrix*. For what concerns our treatment in this Section, we won't go through such a formalism and describe this case by the simpler observation that the coefficients are not well determined.
To quantify the performance of the single photon source, with reference to an idealized device, it is customary to consider the following properties.

Single-photon purity Is the property of the source of producing pulses that contain no more than one photon. A high single-photon purity means that $c_{l,n} \simeq 0$ for all l when $n > 1$. The $g^{(2)}(\tau)$ quantum coherence function is a good measure for the single-photon purity, as an ideal single-photon source features $g^{(2)}(\tau) = 0$.

Indistinguishability Is the property of the source of producing pulses of light in a pure quantum state, described by a well-determined and fixed combination of coefficients $c_{l,n}$. Single-photon pulses emitted at two different times instants should not be distinguishable from one another. We discuss in the next Section a standard way to quantify the indistinguishability of two single-photon states.

Brightness Is the property of the source of producing single-photon pulses in a deterministic way or at least efficiently. A good brightness corresponds to a low amplitude of the coefficients of the

vacuum states $c_{l,0} \simeq 0$ for all l. A possible way to quantify brightness is through the probability that each pulse contains a single photon:

$$\mathcal{B} = \sum_l |c_{l,1}|^2 \tag{9-55}$$

Brigthness is indeed fragile, as any optical loss in the apparatus brings about an increase in the vacuum components.

An **ideal single photon source** should produce pulses in a well determined pure quantum state:

$$|\psi\rangle = \sum_l c_{l,1} |1\rangle_l \tag{9-56}$$

where $\sum_l |c_{l,1}|^2 = 1$ (i.e. unit brightness according to the former definition). Such an ideal source, however, is hardly obtained in the experiments. Practical sources emit states with non-vanishing coefficients for $n \neq 1$. However, it is important to note that, for most applications, the presence of spurious states $|2\rangle, |3\rangle, \ldots$ is far more detrimental than the presence of significant coefficients for the vacuum states. Namely, a low brightness can be more tolerable than a low purity.
To manage the presence of vacuum states in the experiments one proceed typically as follows, adopting a *post-selection* approach. Photon-counting is performed at the output of the experimental setup; only the events for which a single photon, or the desired number of photons in case multiple sources are employed, are detected are considered. If no photons are counted, or if a number of photons different from the ideal one reaches the detector, the result is discarded and the experiment is repeated with new pulses from the sources. If it can be assumed that high-photon number terms ($|2\rangle, |3\rangle, \ldots$) are absent, then counting the correct number of photons is a witness that the experiment has been carried out correctly and, in that instance, no photon has been lost.

9.5 Characterization of photon indistinguishability

Ideally, a good single-photon source should provide subsequent single-photon states that are all identical. If we consider two pure states of a single-photon $|\psi\rangle$ and $|\phi\rangle$, we can define their overlap r as:

$$\boxed{r = |\langle\psi|\phi\rangle|^2 = |\langle\phi|\psi\rangle|^2} \tag{9-57}$$

This overlap takes the maximum value $r = 1$ if the two states are identical, namely **indistinguishable**, and the minimum value $r = 0$ if the two states are orthogonal, namely fully distinguishable.
In detail, the two states can be written as:

$$|\phi\rangle = \sum_l \alpha_l |1_l\rangle \qquad\qquad |\psi\rangle = \sum_l \beta_l |1_l\rangle \tag{9-58}$$

where l is the mode index, $|1_l\rangle$ is the Fock state with $n = 1$ of the l-th mode, and α_l and β_l are complex coefficients. The index l allows to scan all the modes of the chosen basis, with their spatial shape and polarization $\vec{f}(\vec{r})$, as well as frequency ω.
To describe common experimental situation a basis of infinite plane waves might not be the most convenient to take (though always possible, in principle). It could be more practical to consider a basis of *temporal mode*, where each mode is also characterized by a temporal localization. In fact, each of these modes is a linear combination of waves with different ω.[4] We do not need, however, to discuss the details of this basis for what follows.
The overlap r can be estimated experimentally by exploiting the Hong-Ou-Mandel effect. The two single-photon states are injected respectively in the two inputs of a balanced beam splitter, and the

[4] It is indeed well known, from Fourier analysis, that a signal which possesses limited duration in time is obtained from the sum of waves with different frequencies

coincident detection of one photon per each output mode is checked at the output (see Fig. 2). It can be shown[5] that the coincidence detection probability in this case is:

$$p_{11} = \frac{1-r}{2} \tag{9-59}$$

Indeed if the photons are perfectly indistinguishable ($r = 1$) the probability of coincidence detection is totally suppressed by quantum interference (as shown in Section 6.2). Instead, if the photons are fully distinguishable ($r = 0$) we recover the classical probability $p_{11}^{\text{dist}} = 1/2$ that two particles, each having a 1/2 probability of being transmitted in the same mode and a 1/2 probability of being reflected in the other mode, are found one per output mode.

In the experimental practice, the single photon states that are to be compared (from two different sources, or from two separated emission events of the same source) are injected repeatedly in the beam splitter and the actual coincidence detection events at the output are counted in a given time interval. The probability p_{11}, and thus the overlap r from the Eq. (9-59) can be estimated as the coincidence counts are proportional to it. In fact, however, one needs to know the proportionality factor between counts and probability, which depends not only on the emission rate of the sources, but also on the efficiencies and loss contributions present in the apparatus. This is done by introducing in the apparatus a way to add a controlled amount of distinguishability between the two states. Often, this is performed by adding a tunable optical delay line on one of the input arms. The coincidence counts are thus evaluated as a function of the delay τ introduced.

If the delay line is set in such a way that the two input arms are of equal optical length, we are setting $\tau = 0$ and we are estimating precisely the overlap between the two states apart from the proportionality factor κ:

$$C_{11}(0) = \kappa\, p_{11}(0) = \kappa \frac{1-r}{2} \tag{9-60}$$

If the delay line is set to a very long delay $\tau \to +\infty$, we are introducing a strong temporal distinguishability between the two states and the counts correspond to the one expected in the classical limit.

$$C_{11}(\infty) = \kappa\, p_{11}^{\text{dist}} = \kappa \frac{1}{2} \tag{9-61}$$

By plotting the counts as a function of time $C_{11}(\tau)$ a trend such as the one reproduced in Fig. 3 is typically found for a good source of indistinguishable photons, with a dip in correspondence of $\tau = 0$. The visibility of this Hong-Ou-Mandel dip V_{HOM} is defined as:

$$V_{\text{HOM}} = \frac{C_{11}(\infty) - C_{11}(0)}{C_{11}(\infty)} \tag{9-62}$$

By substituting the previous expressions for C_{11} one readily finds:

$$V_{\text{HOM}} = \frac{\kappa\frac{1}{2} - \kappa\frac{1-r}{2}}{\kappa\frac{1}{2}} = r \tag{9-63}$$

The visibility of the dip is a direct measurement of the indistinghuishability of the two single-photon states.

[5] See e.g. J. C. Garcia-Escartin and P. Chamorro-Posada "SWAP test and Hong-Ou-Mandel effect are equivalent" *Phys. Rev. A* **87**, 052330 (2013).

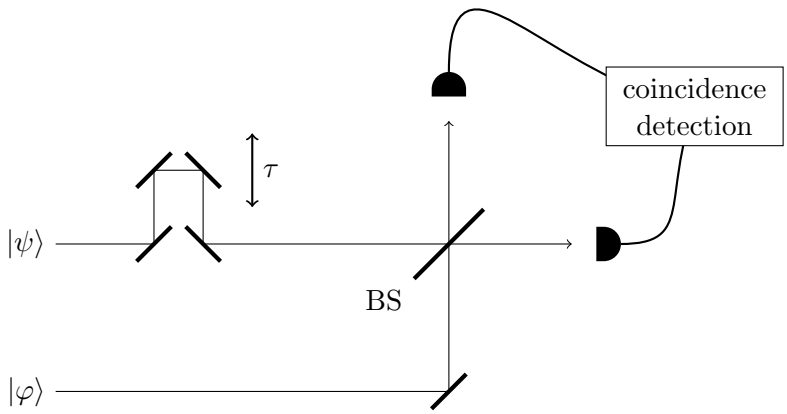

Figure 2: Scheme of an experimental setup to measure the indistinguishability between two single-state $|\psi\rangle$ adn $|\phi\rangle$. The two single-photon states are injected in the two input ports of a balanced beam-splitter (BS), coincidence detection of two photons (one per port) is performed at the output, while scanning the time delay τ. The tunable time delay τ is introduced by changing the length of the optical path of one input arm. When the optical lengths of the two arms (travelled by photons in $|\psi\rangle$ and $|\phi\rangle$) are perfectly balanced, $\tau = 0$; otherwise, a tunable amount of distinguishability is added.

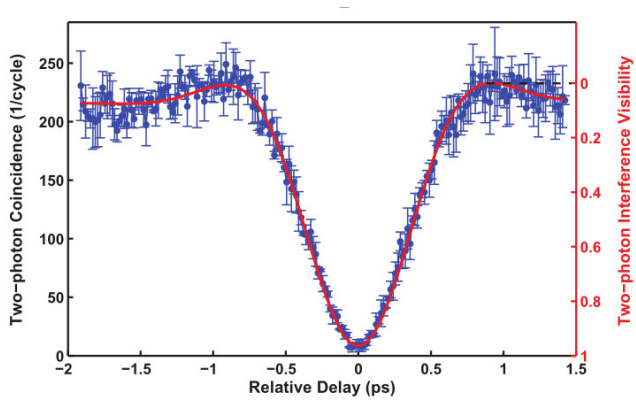

Figure 3: Example of graph plotting the coincidence counts as a function of the relative delay, in a Hong-Ou-Mandel interference experiment involving two highly indistinguishable single-photon states. Image taken from H. Li et al. *New Journal of Physics* **15**, 063017 (2013).

9.6 Types of single-photon sources

At present, different technological approaches are available in order to build a device that approximates the ideal single-photon source operation. Each approach allows for specific advantages, and suffers from some limitations and non-ideality. We discuss briefly, in the following, a few of the techniques that are most commonly adopted to obtain single-photon emitting devices.

Attenuated coherent beams

Strongly attenuating a coherent beam, such as a laser beam, is possibly the simplest approach to build a source of single photons. Indeed, by introducing a sufficient attenuation, the probability to detect more than one photon in a given time bin can be made arbitrarily small. This comes at the price of some relevant drawbacks as we will see shortly.
The photon statistics of the attenuated beam follows the Poisson distribution:

$$p(n) = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$$

the $|\alpha|$ coefficient depends on the optical power but also on the chosen time bin. The probability to have 0, 1 or more photons in a bin reads:

$$p(0) = e^{-|\alpha|^2} \qquad p(1) = e^{-|\alpha|^2}|\alpha|^2 \tag{9-64}$$

$$p(n > 1) = 1 - p(0) - p(1) = 1 - e^{-|\alpha|^2}(1 + |\alpha|^2) \tag{9-65}$$

Therefore, the ratio $p(n > 1)/p(1)$ for $|\alpha| \ll 1$, reads:

$$\frac{p(n > 1)}{p(1)} = \frac{1 - e^{-|\alpha|^2}(1 + |\alpha|^2)}{e^{-|\alpha|^2}|\alpha|^2} \simeq \frac{|\alpha|^2/2 + o(|\alpha|^4)}{|\alpha|^2 - |\alpha|^4/2 + o(|\alpha|^4)} \simeq \frac{|\alpha|^2}{2} \tag{9-66}$$

One observes that this ratio can be made arbitrarily small by sufficiently lowering $|\alpha|^2$, namely by taking a strong enough attenuation. However, one also notes that the brightness decrease proportionally, as:

$$\mathcal{B} = p(1) \simeq |\alpha|^2 \tag{9-67}$$

Therefore, to decrease unwanted multi-photon components one also have to decrease the brightness. A compromise between the two must be reached, in a given applicative scenario.
In addition, and most importantly, notwithstanding the attenuation the beam remains in a quasi-classical state and the authentic statistics of the beam remains Poissonian. As a consequence, its quantum coherence function remains precisely equal to that of a coherent state, with $g^{(2)}(0) = 1$, which does not show genuine single-photon purity.
In fact, due to its simplicity and low cost, attenuated lasers finds use as photon sources in random number generation and quantum cryptography applications, where single photons are used, one at a time, to encode information as qubits. However, $g^{(2)}(0) \neq 0$ impedes to take advantage from more complex quantum interference phenomena between two or more emitted photons (from the same or from distinct sources). This strongly limits the application of these sources to other contexts.

Heralded sources based on photon-pair generation

This kind of sources requires a physical process that generates simultaneously pairs of photons, which may be divided in two optical modes. Detection of one photon in one mode indicates that there will be a photon also in the other modes. This mechanism is called **heralding**. If the probability of producing two pairs simultaneously is sufficiently low, the 'herald' photon is announcing the presence of a single photon in the other mode.
Different phenomena can be exploited to this purpose, such as peculiar fluorescence emission processes that involve two photons, or non-linear optical processes that convert pump photons into pairs of

photons at other wavelengths. To discuss a significant example, we will focus here on **spontaneous parametric down-conversion (SPDC)**, a nonlinear frequency conversion phenomenon which finds wide application in sources of photon pairs. In this process, occurring in a nonlinear optical medium, one pump photon (with frequency ω_P and wavevector $\vec{k}_P$) is converted into two photons called *signal* and *idler*, with respectively frequencies ω_S and ω_I and wavevectors $\vec{k}_S$ and $\vec{k}_I$.
Energy and momentum conservation impose the following relations:

$$\omega_P = \omega_S + \omega_I \qquad \vec{k}_P = \vec{k}_S + \vec{k}_I \tag{9-68}$$

The condition for which momentum conservation is satisfied, for a given pair of frequencies ω_I, ω_S that conserve energy, is called **phase matching**. Note that in an optical medium $k = \omega n/c$ (where n is the refractive index and c is the speed of light in the vacuum) and that the refractive index changes with the frequency $n = n(\omega)$ due to dispersion. Thus, the energy conservation condition does not warrant, by itself, to achieve also phase matching, for three collinear beams with $\omega_P \neq \omega_S, \omega_I$. Possible methods to achieve phase matching include the exploitation of crystal birefringence or, especially in waveguide structures, the use of periodic poling.
The Hamiltonian of the SPDC process is similar to the Hamiltonian of squeezing we have discussed previously, but involving the production of two photons in two possibly different modes (it is called indeed *two-mode squeezing*). It may be written in the following form:

$$\hat{H}_{\text{SPDC}} = -\iota\xi(\hat{a}_S^\dagger \hat{a}_I^\dagger - \hat{a}_S \hat{a}_I) \tag{9-69}$$

where $\hat{a}_{S,I}$ and $\hat{a}_{S,I}^\dagger$ are annihilation and creation operators of the signal and idler modes. The coherent pump beam is considered as a classical wave in this form of $\hat{H}_{\text{SPDC}}$, and its intensity determines the amplitude of the coefficient ξ.
At the output, one finds a state akin to the following:

$$|\psi_{\text{SPDC}}\rangle = \sqrt{1-\lambda^2} \sum_{n=0}^{+\infty} \lambda^n \, |n\rangle_S \, |n\rangle_I \tag{9-70}$$

where $|n\rangle_S \; |n\rangle_I$ are the Fock states in the signal and idler modes and $\lambda < 1$ depends on the pump power and on the efficiency of the process. Note that the output state $|\psi_{\text{SPDC}}\rangle$ is a superposition of Fock states with the same number of photons in the signal and idler modes. The total number of photons in each component is always even, as in the case of the squeezed vacuum.
Examining the statistics of one of the two beams, i.e. the signal or the idler taken alone, one finds:

$$p_S(n) = p_I(n) = (1-\lambda^2)\lambda^{2n} \tag{9-71}$$

This statistics is called *thermal* and is intrinsically super-Poissonian.[6]
As a matter of fact, in the source operation, we are only considering the emission on the signal mode that goes together with a detection on the herald mode. The photon statistics of the signal mode *conditioned on* the possible detection of one photon on the idler mode is indeed significantly different. Neglecting losses and inefficiencies we have, for any $n \geq 1$:[7]

$$p_H(n) = \frac{p_S(n)}{p_I(n>0)} = \frac{(1-\lambda^2)\lambda^{2n}}{(1-\lambda^2)\sum_{m=1}^{+\infty}\lambda^{2m}} = \frac{\lambda^{2n}}{\sum_{m=0}^{+\infty}\lambda^{2m} - 1} = \frac{\lambda^{2n}}{\frac{1}{1-\lambda^2} - 1} = \frac{1-\lambda^2}{\lambda^2}\lambda^{2n} \tag{9-72}$$

[6]For this distribution one can show that

$$\langle n \rangle = \frac{\lambda^2}{1-\lambda^2} \quad \text{Var}(n) = \frac{\lambda^2}{(1-\lambda^2)^2}$$

It is easily noted that $\text{Var}(n) > \langle n \rangle$.

[7]The probability of having n photons on the signal mode, conditioned on the occurrence of a heralding event, is the probability to have both events (heralding *and* n photons in signal modes) divided by the probability of the heralding event. The probability to have n photons on signal *and* heralding is simply $p_S(n)$ for $n \geq 1$, because a state with at least one photon in the signal mode has also at least one photon in the idler (herald) mode. The probability of having some heralding is $p_I(n > 0)$.

while, in principle, $p_H(0) = 0$.
Note that the ratio between $p_H(n > 1)$ and $p_H(1)$ is small and becomes even smaller by decreasing λ:

$$\frac{p_H(n > 1)}{p_H(1)} \simeq \frac{p_H(2)}{p_H(1)} = \lambda^2 \tag{9-73}$$

This allows to achieve an effective single-photon purity through the heralding process.
Unfortunately, as in the case of the attenuated laser, the brightness of this heralded source is linearly related to the former ratio. In fact:

$$\mathcal{B} = p_S(1) = (1 - \lambda^2)\lambda^2 \simeq \lambda^2 \tag{9-74}$$

Obtaining a more pure single-photon state goes together with a lower brightness.

Sources based on single quantum emitters

These sources harness the spontaneous emission from quantum systems that can be modelled with good approximation as two-level atoms.
The photon generation process involves two steps: first, a single quantum system of this kind is excited, optically or electrically; then, the system decays by spontaneous emission. The nature of this two-level system forces the decay to occur only through the emission of a single photon. Emission of more than one photon is indeed basically impossible as far as only a single quantum system is addressed and as far as it can be modelled as a two-level atom.
Importantly, in this kind of sources the single photon purity ($g^{(2)}$) is decoupled from brightness, allowing in principle to overcome the limitations of the approaches described above.
In the past decades, single-photon emission has been demonstrated from different types of quantum systems. In particular, **semiconductor quantum dots** can be reliably modelled as artificial two-level atoms and are a promising choice for use as single photon source.
Quantum dots are nanometric structures that present well defined and well separated energy levels. Their energy structure is basically that of a three-dimensional quantum well. Semiconductor quantum dots can be realized via lithographic techniques. To facilitate emission in the desired optical modes, they are often enclosed in suitably designed optical cavities, realized as microstructures on the semiconductor substrate.